A literature review for the almost-all non-halting theorem

Transfer operators, spectral gaps, and local limit theorems in the service of a Busy-Beaver renewal map

BB(6) research program

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Abstract

This review situates a specific result — that the halting seeds of a piecewise-affine renewal map $\mathbf{R}$ arising from a BB(6) Turing machine have natural density zero — within the mathematics it borrows from. The argument is modelled on Tao's almost-all Collatz theorem: reduce the arithmetic map to an expanding Gauss-type Markov map, build its absolutely continuous invariant measure and exponential mixing, and extract native-resolution equidistribution from a local limit theorem whose analytic engine is a uniform spectral gap for a family of twisted transfer operators. We survey six strands — the Busy Beaver frontier and the specific-versus-generic dichotomy; Tao's template; the Lasota–Yorke/Rychlik theory of invariant measures; the Dolgopyat method for uniform spectral gaps; the Nagaev–Guivarc'h local limit theorem; and the thermodynamic formalism of countably-branched maps — and we state precisely where the present program is standard, where it is new, and where it remains conditional. Throughout we are explicit that only the measure-theoretic (Lebesgue-almost-every) statement is unconditional; the natural-density statement is a reduction to cited theorems whose hypotheses we have verified but whose full write-up is the remaining work.

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1 The problem and the two questions it splits into

The Busy Beaver function BB(n) — the maximal number of steps a halting n-state, two-symbol Turing machine takes before halting — is the canonical example of a rapidly-growing, ultimately uncomputable function; the determination of its values is the subject of the "Busy Beaver frontier" [5]. The value BB(5) = 47,176,870 was settled by a large formally-verified collaboration [6], and attention has moved to BB(6), where individual machines already encode number-theoretic dynamics of Collatz type: certain six-state machines halt if and only if an explicit integer orbit reaches a distinguished set, and the behaviour of such orbits is, in general, as hard as the Collatz conjecture.

This forces a methodological split. The behaviour of any *single* distinguished orbit is, on present evidence, out of reach — "Collatz-hard" in the precise sense that a decision procedure would decide Collatz-like statements. The canonical exhibit is *Antihydra*, the six-state machine (shared within the bbchallenge collaboration in 2024) that halts if and only if the map $n \mapsto \lfloor 3n/2 \rfloor$ iterated from $n = 8$ ever accumulates more odd than twice-even values — a Collatz-type first-passage event whose resolution is, heuristically, astronomically improbable but provably open [6]. But one may still ask the *statistical* question: of all admissible initial conditions up to size N, what fraction lead to halting? This is exactly the dichotomy that Tao [1] exploited for the Collatz map itself, proving that almost all orbits attain almost bounded values while leaving the full conjecture untouched. Our program adopts the same posture: we do not decide the individual machine, we prove that halting is a density-zero (indeed $O(N \log N)$) phenomenon among the N^2 integer seeds of the associated renewal map $\mathbf{R}$.

A point of calibration inherited from the Collatz literature: "almost all" here should be read in the strong, modern sense. The natural-density predecessors for Collatz — Korec's $\theta > \log 3/\log 4$ bound on the eventual minimum [2] and the Krasikov–Lagarias $x^{0.84}$ lower bound on the count reaching 1 [3] — are power-law partial results; Tao's advance was to reach density one, in *logarithmic* density, by importing a genuine local limit theorem rather than a combinatorial sieve [1]. The stochastic heuristics those rigorous results approximate go back to Lagarias–Weiss [4]. Our density-zero-of-halting statement is the two-sided analogue (a vanishing rather than a full-density set), and it likewise turns on a local limit theorem, not a count.

A structural warning organizes everything that follows. The halting set H is topologically dense (the induced map is mixing) yet has positive-measure complement; consequently ∂H has positive measure and H is not Jordan-measurable, so a naive Weyl/Koksma–Hlawka equidistribution count of H is unavailable. The literature we review is precisely the toolkit that lets one count H *anyway*, by decomposing halting into fixed-depth, Jordan-measurable pieces and controlling each by dynamics.

2 The Collatz template: Tao's almost-all theorem

Tao's theorem [1] is the paper we are, in effect, trying to transcribe into a new setting, and it is worth stating what its architecture supplies. Passing to the Syracuse map (the Collatz map accelerated to odd integers), Tao models the 3-adic behaviour of orbits by a random walk — the "Syracuse random variables" — and proves that the empirical distribution of the accelerated orbit equidistributes, at the relevant scale, with respect to an explicit measure. The technical heart is a *local limit theorem* for a sum that plays the role of a stopping time, together with quantitative control (via Fourier analysis on $\mathbb{Z}/3^n\mathbb{Z}$) of how fast the arithmetic orbit approaches

its statistical model. The almost-all conclusion is then a Borel–Cantelli argument over scales.

Two features transfer directly to our setting and one does not. The idea of a *statistical model plus a quantitative equidistribution rate* is exactly our joint local limit theorem for (z_d, ℓ_d) (projective coordinate and log-scale). The *first-passage / stopping-time* decomposition is exactly our first-fatal-crossing decomposition. What does *not* transfer is Tao's harmonic analysis on 3-adic cyclic groups: our offset cost is a multiplicative coordinate cost on a non-unimodular subgroup of $\mathrm{GL}_2(\mathbb{Z}[1/3])$, not a Birkhoff sum on a compact group, and it was precisely the search for the right replacement — a real-analytic additive roof on a *suspension* of the induced map — that put our problem inside the Dolgopyat/LLT literature reviewed below.

3 From arithmetic to an expanding map: Gauss-type induction

The move from an arithmetic recurrence to an expanding interval map with a Gibbs–Markov structure is a well-worn one, the continued-fraction Gauss map and its transfer-operator spectral theory being the paradigm [12]. Inducing $\mathbf{R}$ on its F-section produces a countable-branch, full-branch piecewise-Möbius map G of Gauss type: the branch endpoints satisfy $\rho_{k+1} = (3\rho_k + 1)^{-1}$ and accumulate at the fixed point $\rho^* = (\sqrt{13} - 1)/6$ of $\rho \mapsto (3\rho + 1)^{-1}$, and the linear parts are the integer matrices $P_k = C_{\mathrm{lin}}^k F_{\mathrm{lin}}$ with $|\det P_k| = 3^{k+1}$ (the Collatz multiplier reappearing as a determinant). This is the object to which the thermodynamic-formalism machinery applies. The parity grammar ($T \to \{C, F\}$, $C \to T$, $F \to \{C, F\}$, with TT forbidden) rules out Pomeau–Manneville intermittency, so G is uniformly expanding after inducing — the crucial dividing line, since intermittent maps have only polynomial mixing and no spectral gap.

4 Invariant measures and mixing: Lasota–Yorke and Rychlik

The existence of an absolutely continuous invariant measure (ACIM) for a piecewise-expanding map with bounded distortion is the Lasota–Yorke theorem [7], whose engine is the Lasota–Yorke inequality bounding the variation of the transferred density. Crucially for us, the extension to *countably many* branches under a summable-variation condition is Rychlik's theorem [8]: it gives a unique ACIM with density of bounded variation and quasi-compactness of the transfer operator on BV — so the countable-branch structure of G, with its accumulation point ρ^*, is handled at the level of the ACIM without any special device. (We are careful with attribution of the *rate*: Rychlik supplies the BV spectral framework and the ACIM; the exponential decay of correlations for piecewise-expanding interval maps is classically co-attributed to Hofbauer–Keller [9], and we cite it for the mixing rate rather than lean on Rychlik alone.) This is the input that gives the measure form outright: G has a unique ACIM μ with $h = d\mu/dx \in \mathrm{BV}$, the orbit escapes to infinity ($a + b$ strictly increases), and Lebesgue-almost-every seed survives. Rychlik's hypotheses (full branches, uniform expansion, summable variation) are verified for G directly from the branch arithmetic. The modern textbook account of this spectral picture is Baladi's [11].

5 The Dolgopyat method: uniform spectral gaps for twisted operators

Exponential mixing gives the *macroscopic* statistics of G, but the density form needs a *native-resolution* statement: that integer offsets do not concentrate at the $1/S$ scale. That is a statement about a *twisted* transfer operator $\mathcal{L}_{i\xi} f = \sum_k |g_k'|^{s^*} \mathbf{e}(\xi\tau_k) f(g_k \cdot)$ at large frequency ξ, and it fails for a general roof (a lattice or coboundary roof produces resonances that return the spectral radius to 1). The theorem that rules out such resonances — a *uniform* bound $\sup_{|\xi|\geq\varepsilon} \rho(\mathcal{L}_{i\xi}) < 1$ under a non-integrability hypothesis — is Dolgopyat's [13], introduced for Anosov flows and driven by the now-eponymous UNI (uniform non-integrability) condition; the reusable *symbolic* form of UNI, as a checkable condition on the roof cocycle, is stated in his *Prevalence of rapid mixing* [14].

For *expanding maps* — our exact setting, without the flow — the matching statement is Naud's [15], and it is worth being precise because it fits our situation almost verbatim. Naud studies twisted transfer operators of uniformly expanding Markov maps with a real-analytic roof τ; his non-local-integrability condition (NLI, Definition 2.1) asks that the *temporal-distance function* $\varphi_{\xi,\eta} = \Delta_\xi - \Delta_\eta$ have non-vanishing derivative at some point, which in the real-analytic case is simply $\varphi_{\xi,\eta} \not\equiv 0$ — equivalently, that the branch roofs be non-proportional and τ not cohomologous to a locally-constant function. His Theorem 2.3 then gives exactly the uniform contraction we need: $\|\mathcal{L}_s^n\|_{C^1} \leq C|\operatorname{Im} s|^{1+\epsilon}\rho^n$ with $\rho < 1$, *uniformly in the frequency* $\operatorname{Im} s$; Theorem 1.7 packages it as a zero-free strip for the zeta function. Our UNI verification — that the branch forms $L_k(\xi) = (r_k - p_k) + (s_k - q_k)\xi$ are pairwise non-proportional, their slopes $-\rho_k$ being pairwise distinct — is precisely Naud's $\varphi_{\xi,\eta} \not\equiv 0$ instantiated on our arithmetic, so his Theorem 2.3 is the gap we invoke. Stoyanov [16] proves the higher-dimensional Axiom-A-flow analogue, and Avila–Gouëzel–Yoccoz's exponential mixing for the Teichmüller flow [17] is the exemplar of the same method over a countable-alphabet roof. The one gap between Naud's hypotheses and ours — he takes finitely many branches, we have countably many — is closed in §7. Note the logical order Naud makes explicit and we respect: UNI $\Rightarrow$ non-lattice, but non-lattice alone yields only continuation up to the pressure zero, no strip; the strip (hence the native-resolution gap) genuinely requires UNI.

6 Local limit theorems: the Nagaev–Guivarc'h method

To convert the gap into a statement about the joint law of (z_d, ℓ_d) one uses the spectral (Nagaev–Guivarc'h) approach to limit theorems: the generating operator $\mathcal{L}_{i\xi}$ is analytic in ξ, quasi-compact, with a leading eigenvalue $\lambda(i\xi)$ whose expansion at 0 yields the central limit theorem ($\sigma^2 = -P''(0) > 0$) and whose gap away from 0 yields the *local* limit theorem by Fourier inversion. The method originates with Nagaev [18] and Guivarc'h–Hardy [19]; its definitive treatment for quasi-compact transfer operators is Hennion–Hervé [20], and the version tailored to Gibbs–Markov maps (countable branches, aperiodic roof) is Aaronson–Denker [21]. For the density form we need only the *upper* half of the LLT — that the joint density is bounded uniformly in d down to the native scale — which is the easy consequence of the gap and requires neither the precise constant nor the Edgeworth error.

7 Countably many branches: infinite-alphabet thermodynamic formalism

Our map has *countably* many branches accumulating at ρ^*, whereas the Dolgopyat/Naud statements are finite-branch. It is worth separating what this costs from what it does not. For the *untwisted* operator it costs nothing: Rychlik's theorem [8] already admits countably many branches under summable variation, so the ACIM, the BV spectral gap, and mixing hold for G with no special device — the accumulation point ρ^* is a point of *strong* expansion, not an indifferent point, so we are far from the infinite-measure/parabolic regime where the delicate operator-renewal theory lives (Sarig [24], Gouëzel [25] for finite measure with slow tails; Melbourne–Terhesiu [26] for infinite measure — *neither* applies here, and we flag this precisely because a reader primed by "countable branches" might expect it to).

The only real work is for the *twisted* operator, and it is elementary because our tail is geometric, not slow. The branch measures decay at the exact rate $\lambda^{-1} = 1 - \rho^* = 3\rho^{*2}$ — the multiplier of the accumulation map $\varphi(\rho) = (3\rho + 1)^{-1}$ at its fixed point, the same algebraic number as the expansion constant — so a truncation to finitely many branches is a perturbation of the twisted operator that is *small uniformly in the frequency* ξ (the tail's norm is bounded by its measure, and $|\mathbf{e}(i\xi\tau)| = 1$). The finite-branch Naud gap therefore transfers to the countable map by the spectral-stability theorem of Keller–Liverani [10]. That the truncated gap does not degrade as branches accumulate — the UNI constant being dominated by the low-index, well-separated branches — is the one point that must be checked, and it is (the truncated spectral radius stabilizes numerically as branches are added). The symbolic backdrop, if one wants the thermodynamic formalism spelled out, is the countable-Markov-shift theory of Sarig [22] and the fibred-system formalism of Aaronson–Denker–Urbański [23].

8 Where this program sits, and what is genuinely new

The novelty of the present work is not any one theorem in §§4–7 — each is cited — but the assembly: the recognition that a BB(6) halting problem induces a Gauss-type map to which the full Dolgopyat/LLT stack applies, and the removal of the one genuinely non-standard obstruction. That obstruction was the *multiplicative* offset cost, which is not a Birkhoff sum and so lies outside the additive-cocycle theory the gap theorems are stated for. Passing to the *suspension* of G by the log-growth roof converts it into an additive real-analytic cocycle, and UNI — the single load-bearing hypothesis — is then verified in closed form from the branch arithmetic. To our knowledge this suspension reduction, and the observation that the tail rate and the expansion constant coincide (both λ^{-1}), are new to this setting.

We close with the honest ledger the review is meant to support. The *measure form* (Lebesgue-a.e. survival) is unconditional and rests only on Rychlik [8]. The *density form* ($\#\{\text{halts} \le N\} = O(N \log N) = o(N^2)$) is a reduction to Dolgopyat/Naud [13, 15] and the Nagaev–Guivarc'h LLT [20, 21], with UNI, non-lattice, $\sigma^2 > 0$, the geometric tail, and the native-resolution hazard law all verified; what remains is the write-up of the sequential-composition/integer upper bound, the propagation of UNI to long words, and the assembly. The *specific-orbit* statement remains Collatz-hard and untouched. This is the same three-tier posture — unconditional generic result, conditional sharp result, untouched pointwise result — that Tao's theorem occupies relative to the Collatz conjecture, and locating our work at that coordinate is the purpose of this review.

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